

PAPER_173: Modular Compressed MUGE — 9-Term Mathematical Decomposition

Abstract

This paper presents a UQFF analysis of astrophysical observables, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

Whitepaper §2.4-E | Thread 381a8fe7 | Session 48

Abstract

The Modular Unified Gravity Equation (MUGE) in compressed form decomposes gravitational dynamics into 9 independent sub-terms. Each term captures a distinct physical contribution: Newtonian base, cosmological expansion, magnetic suppression, environmental context, Ug contributions, cosmological constant, quantum corrections, fluid dynamics, and density perturbations.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$g_{\text{UQFF}}(r) = g_{\text{MUGE}}(r) \cdot \left(1 - [SSq] \cdot U_{b_i} / F_U(r, t)\right), \quad [SSq] = 0.57$$

1. MUGESystem Struct

```
struct MUGESystem {
    std::string name;
    double I;           // Moment of inertia [kg·m²]
    double A;           // Area / cross-section [m²]
    double omega1, omega2; // Angular frequencies [rad/s]
    double Vsys;        // System volume [m³]
    double vexp;        // Expansion velocity [m/s]
    double t;           // System age [s]
    double z;           // Redshift
    double ffluid;      // Fluid frequency [Hz]
    double M;           // System mass [kg]
    double r;           // Characteristic radius [m]
    double B, Bcrit;    // Magnetic field and critical field [T]
    double rho_fluid;    // Fluid density [kg/m³]
    double g_local;     // Local gravitational acceleration [m/s²]
    double M_DM;        // Dark matter mass [kg]
    double delta_rho_rho; // Density perturbation ratio
};
```

2. Nine Sub-Terms

Term 1 — Newtonian Gravitational Baseline

$$\text{compressed_base} = G \times M / r^2$$

Constants: $G = 6.67430\text{e-}11$

Test: $M=1.989\text{e}30$ kg, $r=1.496\text{e}11$ m ? ~ 0.0059 m/s²

Term 2 — Hubble Expansion

$$\text{compressed_expansion} = 1 + H_0 \times v_{\text{exp}}$$

$H_0 = 2.269\text{e-}18$ s?¹ (~ 70.1 km/s/Mpc)

At $t=0$: expansion = 1.0

Term 3 — Magnetic Suppression (Super Adjustment)

$$\text{compressed_super_adj} = 1 - B/B_{\text{crit}}$$

Test: $B=1\text{e}10$ T, $B_{\text{crit}}=1\text{e}11$ T ? 0.9

Above B_{crit} : approaches 0 (magnetic quench)

Term 4 — Environmental Factor

$$\text{compressed_env} = 1.0 \quad [\text{placeholder for ISM/nebular environment}]$$

Term 5 — Ug Contributions Sum

$$\text{compressed_Ug_sum} = 0.0 \quad [\text{Ug interface placeholder for future coupling}]$$

Term 6 — Cosmological Constant Term

$$\text{compressed_cosm} = ? \times c^2 / 3$$

? = $1.1\text{e-}52$ m?² (dark energy)

$c = 3\text{e}8$ m/s

? compressed_cosm = $1.1\text{e-}52 \times 9\text{e}16 / 3 \sim 3.3\text{e-}37$ m/s²

Term 7 — Quantum Correction

$$\text{compressed_quantum} = (? / ?x_p) \times ?? \times (2p / t_{\text{Hubble}})$$

Parameters:

$$? = 1.0546\text{e-}34 \text{ J}\cdot\text{s}$$

$$?x_p = ?x \times ?p = 1\text{e-}68 \text{ J}\cdot\text{m} \text{ (minimal uncertainty product)}$$

$$?? = \text{integral_psi} = 2.176\text{e-}18 \text{ (ground state energy proxy)}$$

$$t_{\text{Hubble}} = 4.35\text{e}17 \text{ s}$$

$$? \text{ quantum} = (1.0546\text{e-}34 / 1\text{e-}68) \times 2.176\text{e-}18 \times (2p / 4.35\text{e}17)$$

$$= 1.0546\text{e}34 \times 2.176\text{e-}18 \times 1.443\text{e-}17$$

$$\sim 3.312\text{e-}1$$

Term 8 — Fluid Dynamics

$$\text{compressed_fluid} = \rho_{\text{fluid}} \times V_{\text{sys}} \times g_{\text{local}}$$

Test (SGR1745): $\rho_{\text{fluid}}=1\text{e-}15$, $V_{\text{sys}}=4.189\text{e}12$, $g_{\text{local}}=10.0$

$$? \text{ compressed_fluid} = 1\text{e-}15 \times 4.189\text{e}12 \times 10 = 4.189\text{e-}2$$

Term 9 — Density Perturbation

$\text{compressed_perturbation} = M \times (\text{delta_rho_rho} + 3 \times G \times M / r^3)$
Captures dark matter and baryonic density contrast effects.

3. Full Compressed MUGE

$\text{compressed_MUGE} = \text{base} \times \text{expansion} \times \text{super_adj} \times \text{env} \times (1 + \text{Ug_sum})$
 $+ \text{cosm} + \text{quantum} + \text{fluid} + \text{perturbation}$

Expected (SGR1745):

base ~ $G \times 2.984e30 / 1e8 \sim 1.99e11$
expansion ~ $1 + 2.269e-18 \times 1e3 \sim 1.0$
fluid = $4.189e-2$
perturbation ~ $M \times d^?$ terms
Total ? ~ $1.782e39$ (from unit test)

4. Canonical System Test Values

System	compressed_MUGE [m/s ²]
SGR 1745-2900	~ $1.782e39$
Sagittarius A*	~ $1.782e39 \times (M_{\text{SgrA}} / M_{\text{SGR}})$
Student Guide	cosmological scale

5. Relationship to SOURCE4

The 9-term compressed MUGE directly corresponds to the `compute_compressed_MUGE_SOURCE4()` function in `MAIN_1_CoAnQi.cpp` (lines 25623–26026, namespace `SOURCE4`). The thread 381a8fe7 version provides the modular sub-term decomposition enabling independent validation.

6. References

- MUGE.h/cpp (thread 381a8fe7)
- UnitTests.cpp lines 1–200 (test_compute_compressed_MUGE expected= $1.782e39$)
- PAPER_174 (resonance MUGE, same MUGESystem struct)
- SOURCE4 integration commit 3e66d94